TRACHEA

Regular anatomy notes - definition, extent, relations, supply, structure and applied anatomy

Core idea

The **trachea** or windpipe is a flexible fibrocartilaginous air tube that forms the beginning of the lower respiratory tract.

Its lumen remains open because of **16-20 C-shaped hyaline cartilaginous rings**. The posterior gap is completed by trachealis muscle and fibroelastic tissue, allowing the esophagus to expand during swallowing.

In revision, remember the trachea as a tube with three exam pillars: **extent and dimensions**, **relations**, and **applied anatomy**.

High-yield facts

Extent: lower border of cricoid cartilage at C6 to lower border of T4, where it bifurcates.

Parts: cervical part in neck and thoracic part in superior mediastinum.

Bifurcation: near sternal angle; forms right and left principal bronchi.

Why the wall does not collapse

Cartilage rings keep the airway patent.

Trachealis muscle narrows the lumen during coughing and helps increase the velocity of expired air.

Elastic tissue allows flexibility during head and neck movements.

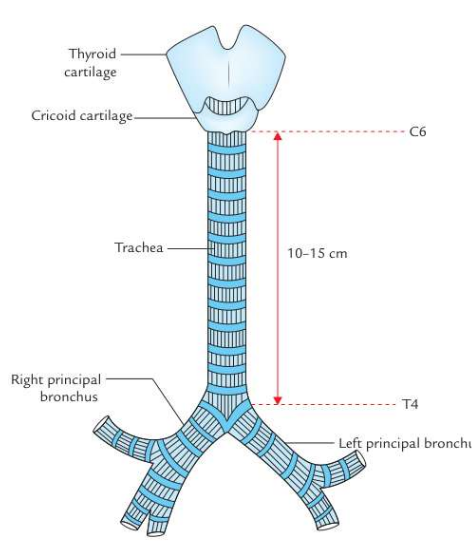


Fig. 22.1 Trachea.

General extent and course

The trachea begins at **C6** as the continuation of the larynx and descends downward and backward.

It enters the thoracic inlet in the midline and ends at **T4**, a little to the right of the midline, by dividing into the two principal bronchi.

The right principal bronchus is wider, shorter and more vertical than the left, so inhaled foreign bodies more commonly enter the right bronchial tree.

1. Location, extent and dimensions

Feature	Important points

Feature	Important points
Location	Upper half lies in the neck as the **cervical trachea**; lower half lies in the **superior mediastinum** as the thoracic trachea.
Extent in cadaver	**C6 to T4** in supine position.
Extent in living standing adult	May reach **C6 to T6** because of posture and descent of thoracic structures.
Extent in newborn	About **C6 to T3**, so the trachea is relatively short.
Length	About **10-12 cm**.
External diameter	About **2 cm in males** and **1.5 cm in females**.
Internal diameter	About **12 mm in adult** and **3 mm in newborn**.

Child airway note

In childhood, the internal diameter approximately corresponds to age in years until it reaches adult size. This is why endotracheal tube size is expressed in millimetres and must be chosen carefully in children.

A small reduction in the radius of a child airway can markedly increase airway resistance, making edema or obstruction clinically serious.

2. Course

Course in one paragraph

The trachea is the direct continuation of the larynx. It starts at the lower border of the cricoid cartilage opposite the lower border of C6 vertebra, about 5 cm above the jugular notch. It passes downward and backward behind the manubrium and terminates at the lower border of T4 vertebra by bifurcating into the right and left principal bronchi. The bifurcation corresponds approximately to the sternal angle and is slightly to the right of the median plane.

Revision line

Cricoid cartilage - **C6**.

Sternal angle - **T4/T5 plane**.

Tracheal bifurcation - near **lower border of T4**.

Palpation point

The trachea can be palpated in the **suprasternal notch**.

Normally it is central. Appreciable deviation suggests **mediastinal shift** or displacement by a neck/thoracic lesion.

3. Structure of the tracheal wall

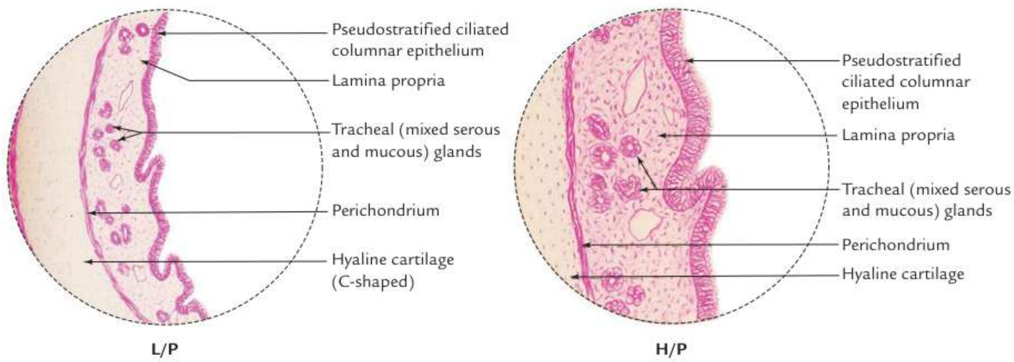


Fig. 22.4 Microscopic structure of trachea (L/P = low power, H/P = high power). (*Source:* Box 16.2, Page 349, *Textbook of Histology and a Practical Guide*, 2e, JP Gunasegaran. Copyright Elsevier 2010, All rights reserved.)

Source diagram: microscopic structure of trachea showing respiratory epithelium, glands, perichondrium and hyaline cartilage.

Layer	Description and importance
Mucosa	Lined by **pseudostratified ciliated columnar epithelium** with goblet cells. Cilia move mucus upward toward the pharynx.
Lamina propria	Contains elastic fibres; gives support and elasticity to the mucosa.
Submucosa	Loose areolar tissue with numerous **serous and mucous glands**. These glands help moisten and protect the airway.
Cartilage and smooth muscle layer	Horseshoe-shaped **C-shaped hyaline cartilage rings** are incomplete posteriorly. Posterior gap contains trachealis muscle and fibroelastic fibres.
Perichondrium	Surrounds and nourishes the hyaline cartilage.
Fibrous membrane	Dense connective tissue containing neurovascular structures.

Functional interpretation of the wall

The trachea needs to be **rigid enough** to remain open but **flexible enough** to move with swallowing, breathing, and movement of the neck.

The C-shaped cartilage supports the anterior and lateral walls, while the posterior membranous wall allows the esophagus behind it to expand during passage of a food bolus.

4. Cervical part of trachea

Extent and position

The cervical part is about **7 cm long** and extends from the lower border of the cricoid cartilage to the upper border of the manubrium sterni.

It descends downward and slightly backward in front of the esophagus, following the curvature of the cervical spine. It enters the thoracic cavity near the median plane with a slight deviation to the right.

Relation	Structures
Anterior	Skin; superficial fascia with anterior jugular veins and jugular venous arch; investing layer of deep cervical fascia; sternohyoid and sternothyroid; thyroid isthmus over 2nd-4th tracheal rings; inferior thyroid veins; occasional thyroidea ima artery. In children, thymus and left brachiocephalic vein may rise into the neck.
Posterior	**Esophagus** and recurrent laryngeal nerves in the tracheoesophageal grooves.
Right and left sides	Lobes of thyroid gland extending down to 5th or 6th tracheal ring; common carotid artery within the carotid sheath.

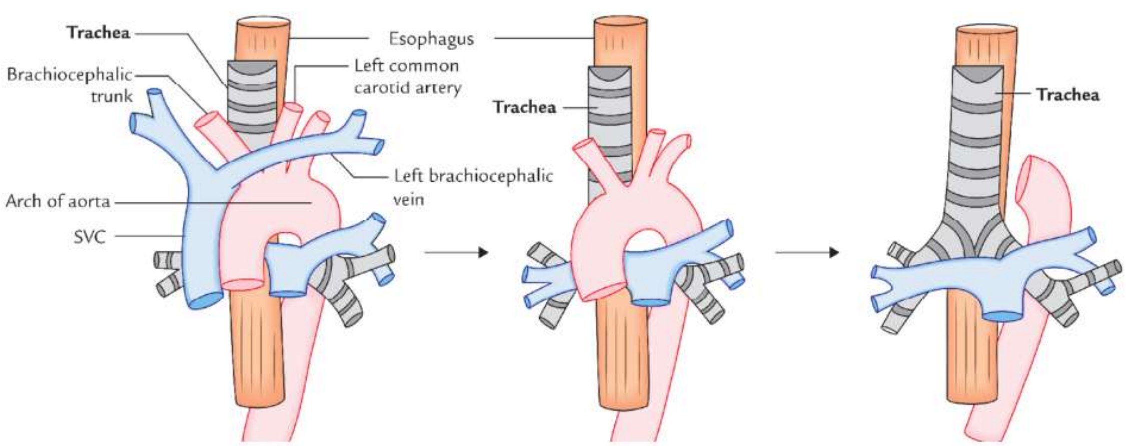
Clinical meaning of cervical relations

The thyroid isthmus is directly anterior to upper tracheal rings, so it must be displaced or divided carefully during tracheostomy.

The recurrent laryngeal nerves lie in the tracheoesophageal grooves; damage to them can affect the vocal cords and voice.

In children, low tracheostomy is risky because thymus and large veins may extend higher into the neck.

5. Thoracic part: anterior relations



Source diagram: anterior relations of the trachea arranged from superficial to deep.

Anterior relation	Why it matters
Arch of aorta	Crosses anteriorly in the superior mediastinum and is closely related to the thoracic trachea.
Brachiocephalic trunk and left common carotid artery	Great vessels lie anterior to the trachea; their pulsations and surgical position are important.
Left brachiocephalic vein and upper SVC	Large veins are anterior and can be injured in low neck or superior mediastinal procedures.
Deep cardiac plexus	Autonomic plexus related to the trachea in the superior mediastinum.

How to read the anterior-relation diagram

Think of the superior mediastinum in layers. From anterior to posterior, the main order is: **thymus/remnant - large veins - large arteries - trachea - esophagus**.

This layering explains why venous bleeding can be significant during procedures near the lower cervical trachea and thoracic inlet.

6. Thoracic part: posterior and lateral relations

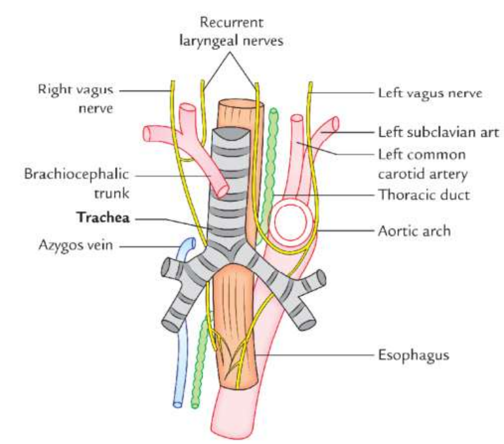


Fig. 22.3 Posterior and lateral relations of the trachea.

Posterior and side relations

Posterior: esophagus, vertebral column and left recurrent laryngeal nerve ascending between trachea and esophagus.

Right side: right lung and pleura, azygos vein and right vagus nerve.

Left side: arch of aorta, left common carotid artery, left subclavian artery, left vagus nerve and left phrenic nerve.

The esophagus lies directly behind the trachea, explaining why the posterior tracheal wall is membranous and expandable.

Side	Key relations
Posterior	Esophagus; vertebral column; left recurrent laryngeal nerve in the tracheoesophageal groove.
Right	Right pleura and lung; azygos vein; right vagus nerve.
Left	Arch of aorta; left common carotid artery; left subclavian artery; left vagus nerve; left phrenic nerve.

Memory hook for side relations

Right side of thoracic trachea is remembered with **RAV**: Right lung/pleura, Azygos vein, Vagus nerve.

Left side is more vascular: arch of aorta, left common carotid artery and left subclavian artery, with left vagus and phrenic nerves.

7. Blood supply, lymphatic drainage and nerve supply

Blood and lymph

Arterial supply: mainly by branches of the **inferior thyroid arteries**.

Venous drainage: into the **left brachiocephalic vein**.

Lymphatic drainage: into **pretracheal and**

Nerve supply

Nerve supply is autonomic.

Parasympathetic fibres are from the vagus through recurrent laryngeal nerves; they are sensory and secretomotor to mucosa and motor to trachealis.

paratracheal lymph nodes.	**Sympathetic fibres** are vasomotor and arise from cervical sympathetic ganglia.

Mucociliary clearance

Mucus traps dust and microorganisms. The cilia of the respiratory epithelium beat upward, moving mucus toward the pharynx, where it can be swallowed or expectorated.

During coughing, trachealis contraction narrows the lumen and increases the velocity of expired air, helping clear secretions from the airway.

8. Applied anatomy

Applied point	Explanation
Tracheal shadow in X-ray	Seen as a vertical translucent shadow in front of the cervicothoracic spine because air is present in the tracheal lumen.
Tracheal palpation	Palpated in the suprasternal notch. Deviation suggests mediastinal shift or displacement by neck/thoracic pathology.
Carina	A keel-like ridge at tracheal bifurcation. It is highly sensitive and commonly initiates cough reflex.
Bronchoscopy	Carina is an important landmark. Distortion or widening of the bronchial angle may suggest enlarged tracheobronchial lymph nodes, including malignant involvement.
Tracheostomy	Life-saving opening of the trachea in laryngeal obstruction. Usually performed after displacing the thyroid isthmus.
High vs low tracheostomy	High tracheostomy is above the isthmus; low tracheostomy is below it. Low tracheostomy is risky in children due to high-positioned thymus and large vessels.

Tracheostomy - key exam paragraph

In tracheostomy, the thyroid isthmus is displaced upward or downward and a vertical incision is made into the trachea, commonly near the 2nd-3rd or 3rd-4th tracheal rings. This is converted into a circular opening for insertion of the tracheostomy tube.

The surgeon must avoid injury to inferior thyroid veins, thyroidea ima artery if present, and high-positioned left brachiocephalic vein or thymus in children.

9. Final revision page

Question	Answer
What is the extent of trachea?	From lower border of cricoid cartilage at **C6** to lower border of **T4**, where it bifurcates.
How many rings support the trachea?	**16-20 C-shaped hyaline cartilaginous rings**.

Question	Answer
Why are rings incomplete posteriorly?	To allow expansion of the esophagus during swallowing and to permit action of trachealis muscle.
Where is the thyroid isthmus related?	In front of **2nd, 3rd and 4th tracheal rings**.
Where are recurrent laryngeal nerves related?	In the **tracheoesophageal grooves**.
Which lymph nodes drain trachea?	**Pretracheal and paratracheal lymph nodes**.
Why is carina important?	It is a sensitive cough-reflex area and an important bronchoscopic landmark.

Do not forget in practicals

Feel trachea in suprasternal notch.
Check whether it is central or deviated.
Identify thyroid isthmus over upper tracheal rings.
Relate trachea posteriorly to esophagus.

One-line summary

Trachea is a **C6 to T4 fibrocartilaginous airway** with C-shaped rings, trachealis muscle posteriorly, thyroid isthmus anteriorly in the neck, esophagus posteriorly, and carina at its bifurcation.

Exam traps: Separate cervical and thoracic relations. Do not forget the thyroid isthmus at 2nd-4th rings, recurrent laryngeal nerves in tracheoesophageal grooves, and carina as the cough-reflex/bronchoscopy landmark.